*Structures and Architecture. REstructure REmaterialize REthink REuse - Rinke & Frier Hvejsel (Eds)*
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# Design for adaptation - Adopting adaption for timber construction at three scale

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ABSTRACT: This paper examines circular construction practices within the timber track of the Eco-Metabolistic Architecture (EMA) project, focusing on reclaimed timber's integration into sustainable architectural workflows. The research conceptualizes *adaptability* as an iterative process linking material properties, contextual considerations, and life cycle histories to support circular flows and cascading reuse. Timber's renewable and recyclable qualities make it suitable for circular construction, but its heterogeneity and transformation over time present challenges. Reclaimed timber presents significant challenges due to its variable density, strength, and structural integrity, coupled with unknown histories and prior damage. Traditional grading systems fail to address these complexities, requiring advanced methods for evaluation. Through three design probes—*ReBeam* (material scale), *Observatory* (element scale), and *ReShelter* (structural scale)—the paper develops adaptive reclamation methods. These approaches redefine design workflows, expanding beyond material attributes to address cascading potential and long-term integration, contributing to sustainable and innovative construction practices.

## 1 INTRODUCTION

Adaptation derives from the Latin "Adaptio", a contraction between tailoring and suitable. As a key concept within evolutionary theory, adaptation describes species' ongoing physical and behavioural adjustments to their surroundings. An ongoing ecological process of change for improving survival rates and lifespans. Building on this understanding, this research paper expands on the theoretical framework of adaptation in circular construction. The paper explores adaptation as an iterative process within circular architecture, operating across three interconnected scales—material, element, and structure. At each scale, adaptation requires different strategies for monitoring, analysis, and evaluation. This framework guides the integration of reclaimed timber into design workflows, supporting sustainable reuse.

In the paper, we use timber construction as a central case study in conceptualising adaptation within circular design. Today, timber is understood as one of the few ecologically sound building materials (Harte 2017) being renewable, recyclable, energy-efficient and having the potential to act as a carbon sink (Ormarsson & Dahlblom 2013). This has led to a renewed interest in timber construction and the emergence of novel methods for its reclamation and repurposing in architectural structures (Ottenhaus et al. 2023). However, reclaimed timber presents unique challenges. Unlike virgin timber, which undergoes standardised grading, reclaimed timber varies significantly due to previous use, environmental exposure and material changes over time. In virgin timber cycles, this material variation is controlled through practices of material grading. Construction timber is classified according to strength grading through visual or machine sorting done at the sawmill. In contrast, reclaimed timber lacks grading standards, and is often characterised by unknown origin, material density, contamination, infestation, currently declassifying defects, including cracks, creep and shifts in the inner morphology through repeated load cycles and natural occurring

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changes on the crystalline level of timber through aging, that result in altered mechanical behaviours (Cavalli et al. 2016).

This paper asks how to ideate a future building practice in which differentiated resource streams are merged. Learning from industrial ecology, it builds a metabolist understanding of material cycles in which the full life cycle of resources is comprehensively understood from resource extraction to cycled deployment across multiple life spans. The paper contributes the outlining of a novel design workflow that incorporates virgin and reclaimed material flows. The process of monitoring, analysis, and evaluation plays a fundamental role in this workflow. Rather than treating these steps as separate processes, they are integrated within the wider design workflow to ensure reclaimed materials are assessed effectively and incorporated seamlessly. Through this the paper provides a new model for how a building practice of diverse resource cycles can be imagined.

## 2 BACKGROUND

Adaptability is at the centre of the circular design framework. The ability to actively re-fit and reconstruct building assemblies lies at the heart of cascading. Adaptability endows buildings with the ability to be changed in response to external impacts such as changing needs, environmental conditions or building regulations. As such, it addresses both the programmatic and spatial dynamics that buildings undergo as they are refitted and refurbished for new use. At the same time, adaptability is used to describe the reversibility of materials through material cascading (Hamida et al. 2023; Giorgi, Lavagna & Campioli 2020). The ability to reclaim and remanufacture building components and repurpose them in new building assemblies is central to the hierarchy of increasingly intense cascades. In this vision, indebted to the contributions of the field of Industrial Ecology, cities are perceived and strategically operated as material banks where building elements are reclaimed, remanufactured and repurposed into new building assemblies and adaptability serves as the method of building incorporating dismantlability, relocate-ability and remanufacture-ability (ibid p 52).

They key to this second resource driven strand of thinking of adaptability in circular design lies with the core ambition of placing buildings and their elements within planned value chains (McKinsey Sustainability 2016). Cascading is fundamentally about retaining the value of artefacts by repurposing the value-creation that design and fabrication offers and therefore implement cascading at the tightest possible loops while at the same time proffering known paths for uptake in future life cycles. Adaptability is therefore both about direct redeployment of existing building elements in new assemblies through modes of disassembly as well as the facility by which the material they represent can be reclaimed and remanufactured.

While these methods are still emerging, they offer profound challenges to the practices of architectural design. Architectural design is founded in methods of engaging with the contexts of environment and site. Designing for adaptability necessitates new methods that can extend these practices to further engage with the past life cycles and past performances of assemblies, elements and materials. Furthermore, the designation of differentiated cascades tends to separate material cycles into distinct practices. The reality is however that these practices are highly interconnected and that a single building might embody multiple simultaneous strategies of adaptation. What emerges is a complex ecology in which adaptability transcends through the scales; from the macro scale of adaptable buildings construed through adaptable building systems, to the meso-scale of adaptable building elements and finally to the micro-scale adaptable materials. At present these inter-scale methods are still only emerging. While a scale-based stratification of cascading has been deployed, this has mainly been understood at a systemic level (Ghisellini, Cialani & Ulgiati 2016). By shifting the focus of inter-scale thinking towards the sources of reclamation, our aim is to ideate the foundational methods of a future circular building practice in which differentiated material streams, both virgin and reclaimed, are matched with appropriate modes of adaptability. Working across the scales of the material, element and assembly adaptability offers flexibility across the design process enabling response to nuanced conditions while planning for change according to processes at work. In this framework, adaptation becomes a method for engaging with building as an iterative and ever-evolving process, a metabolic act in which input streams are aligned with carefully managed output streams, creating a cohesive and sustainable framework occupying diverse timeframes (Figure 1).

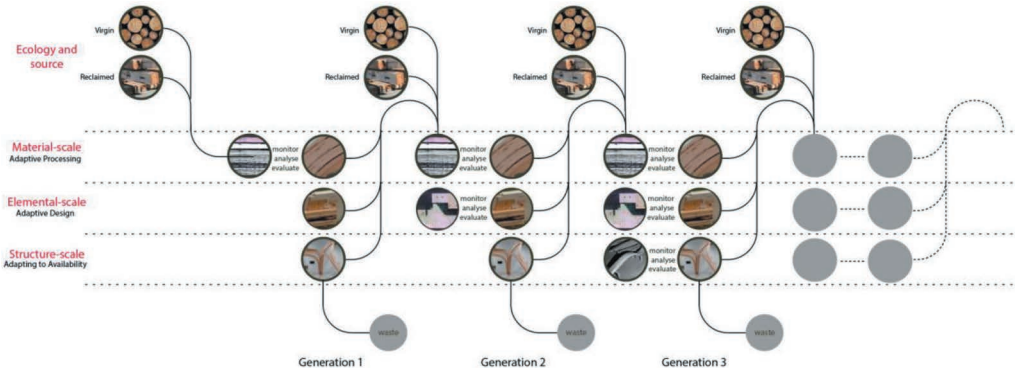



Figure 1. Resource flow and the reoccurring steps of monitoring, analyse and evaluation enabling reuse through adaption over the many instances a material, elements and structure can take. The metabolistic conceptualisation of adaptability across the three scales of material, element and structure position building as an iterative act through which materials flow. The three probes outline the differentiated practices of monitoring, analyse and evaluation across multiple generations of building life span.

## 3 ADAPTABILITY IN TIMBER CONSTRUCTION – CONNECTING THREE SCALES OF MATERIAL ADDRESS

This paper reports on the emerging building practices within the timber track of the Eco-Metabolistic Architecture (EMA) project—a research initiative investigating the architectural and technological implications of biobased materials within circular design practices in the building sector. The research is grounded in an ecological understanding of architecture, where resource value, material fragility, and adaptability are central to design processes (Ramsgaard Thomsen, Miller & Dalager Nielsen s.d.). A central focus within EMA is the development of innovative design methods for integrating reclaimed timber into circular construction. By addressing these three interconnected scales, the study investigates future practices for monitoring, analysing and evaluating resources while integrating design-driven workflows. Central to EMA is the conceptualization of *adaptability*—an iterative process that interweaves material properties, contextual considerations, and life cycle histories to support their integration into circular flows. We define methods of adaptive reclamation through the investigation of three probes – in *ReBeam* the adaptability at the scale of the material, in *Observatory* at the scale of the element and in *ReShelter* at the scale of the structure. This paper brings these methods together to outline a metabolistic approach that can shift design thinking beyond the immediate physical attributes of materials, expanding it to address their cascading potential and alignment with broader circular strategies.

Reclaimed timber offers significant potential in circular construction, yet it presents as a series of challenges that complicate its integration into modern building practices. As a naturally heterogeneous material, timber varies in density, strength, and structural integrity due to its growth patterns, exposure to environmental conditions, and historical use. These variations are further amplified in reclaimed timber, where additional uncertainties arise, such as unknown sourcing histories, prior grading classifications, and the presence of defects like knots, wane, or previous mechanical damage. The lack of standardization and consistent quality in reclaimed timber complicates its evaluation and reuse. Traditional methods of strength grading, often visual or machine-based, are insufficient to address the complexities of assessing aged or previously utilized timber. Established methods lack ways to handle the transformations that circular timber has undergone over its lifetime. Reclaimed timber has typically interacted with the environment over long periods of time, resulting in deformations, cracks, local contaminations and infestations, shifts in the inner morphology through repeated load cycles and even natural occurring changes on the crystalline level of timber through aging, that result in altered mechanical behaviours (Cavalli et al. 2016). Circular construction with reclaimed timber therefore necessitates the development of advanced methods for monitoring, analysing, and evaluating reclaimed timber to ensure its performance.

### 3.1 *Developing adaptability at three scales*

This three-part of monitoring, analysis and evaluation lays the foundation for a new architectural practice which we unfold in the following three sections. This novel practice has its base in an adaptability across scales, sourcing, design and fabrication processes. Adaptation is here ultimately data driven and monitoring is the means to generate this data. Here, monitoring encompasses the tracking and assessing of a material's performance, properties, and condition over time, which can involve observing factors like degradation, structural changes, or contamination. Especially, non-destructive monitoring can determine material properties at a specific point in time, providing data on the material's composition, geometry, mechanical, thermal, or other properties, while allowing the use of the resource post investigation. Monitoring with imaging techniques is highly efficient and techniques like CT scanning can be used in high throughput environments, where data for each individual timber element is generated at high speed. Image based monitoring can provide data in realtime and be deployed on site. It provides data that is tuned to computational image analysis and machine learning that can be crucial future components for the development of individualised adaptation strategies in design and fabrication.



Figure 2. Observatory. Mapping of the found material qualities in timber logs to the required performances (FEA) of individual gluelam beams in the timber structure.

#### 3.1.1 *Material scale – Adaptive processing*

The first probe, ReBeam examines adaptability at material scale. Here, adaptability is understood as the ability to instrumentalise material scale variation (changes in material density, fibre direction and material defects) to compose graded composites in the form of glulam elements. Reclaimed timber has a high degree of heterogeneity and core attributes that help characterise the timber resource can be harder to detect. Where virgin timber uses the detection of pith line and sap to heart wood delineation as a means of deferring assumptions of quality these can be hard to detect in reclaimed timber where this material information is missing. Furthermore, defects such as machining, cracks and nails as well as material deformation due to structural deployment can change the inherent material structure. To design for adaptation at material scale we have developed new means of non-destructive testing that allows an assessment assess the individual properties of each board. The resulting data is tied to the physical entity. In this way we generate a material level digital twin, that can be updated, queried and influence further steps in the processing and circular life of the element.

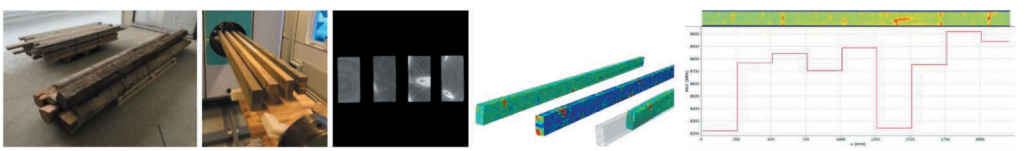


Figure 3. Monitoring, analysis and evaluation of reclaimed planks (l), sample setup during CT scanning (m) a section of the scan (r), full beam FE model to assess the bending stiffness, resulting stiffness curve, that informs the design adaptation.

Monitoring occurs in our project in the form of CT scanning through which material density maps are generated. The density maps are transformed into volumetric models, that describe the materials actual geometry, inner composition, and material differentiation. Finite Element Analysis of this volumetric material representation (material level digital twin) provides a detailed characterisation of each element (Tamke et al. s.d.). In ReBeam, as well as former studies including RAWLAM deploying virgin resource (Tamke et al. 2021; Svilans et al. 2023), this material analysis is interfaced with the design of performance driven Glue Laminated elements. Here, the individual performance of timber elements is matched with local performance needs of the overall design (Figure 2). Instead of assuming a homogenous resource, material scale analysis determines the shape and position of the lamellas in the desined beam. Preliminary results of our research, indicate that the transfer of the method from virgin to reclaimed timber is successful (Figure 3), and our computational methods are able to capture and represent the impact of small-scale defects e.g. cracks and nails, that emerged during the materials life, on its structural performance (Tamke et al. s.d.).

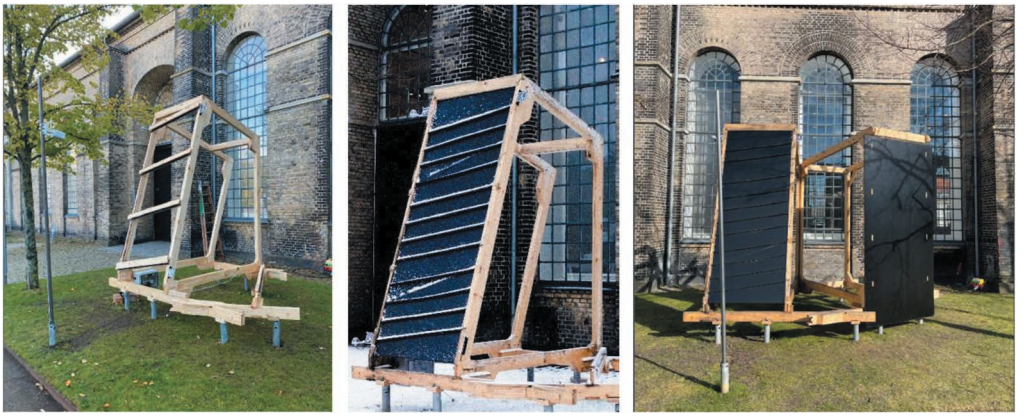


Figure 4. Adaptation and extension of the observatory (from left): initial state, introduction of a board for better support of the biopolymer panels and stiffening of the timber structure, extended structure with a higher biopolymer board in vertical position.

### 3.1.2 *Element scale – adaptive design*

In the second probe, Observatory, a temporary timber structure (Figure 4), we investigate adaptability at element scale. We query how new elements can interface with existing conditions and how adaption can take place within both existing and new tectonic logics. Where timber is in itself a highly malleable material systems that is easy to tool and therefore to adapt, established methods of joining and connecting individual elements are inherently difficult to integrate in circular workflows as they typically deploy non-reversible fastening methods. Adhesives, nails, screws, and other mechanical fasteners can damage the material during removal, reduce its structural integrity and limits its potential for reuse. The excessive use of nails and screws requires costly post-disassembly processing which are predominately manual. This introduces high risk factors to circular processes, as any remaining metal piece is an expensive hazard for timber processing machines and adhesive-bonded laminates often cannot be cleanly separated without destroying the resource.

The Observatory was designed with adaptation and extension in mind. The original structure designed in Spring 2023 was developed as an extendable structural system allowing for future appropriation, supported by a clear and legible structural logic. We incorporate a resilient material logic as elements and joints are over proportioned to take on future loads, the use of reversible timber connections for future disand reassembly and timber elements, which are joined in ways that can be further extended and provide attachment points for additional elements. In this way the Observatory acknowledges the unpredictability of future questions towards the structure and provides means to engage with the through a design for adaptability strategy.

This was tested in Spring 2024 when we extended the structure by 50%. Here, monitoring, in the form of Lidar 3D scanning, is used to develop an element scale digital twin. This enables us to analyse the deformation and changes that occurred in the elements during its first life span. We identify two kinds of deviations: firstly, between design intent and the as-built structure where joints had not been executed as specified in the 3D design files, and secondly, in the actual element geometry which had warped and swelled due to outdoor exposure. The second stage of design was developed to engage with the initial design, where the open joints allowed an easy extension. Precise data on the change and deformation of the elements allowed us to adapt the detailing and shape of the interfaces between existing and new elements to achieve good fitting joints, that did not require laboursome on site adjustments (Figure 5).



Figure 5. Observatory. Initial state of a joint (left) and adaption to the weathered state during the extesnion with new beam elements made from reclaimed timber.

The element scale material representation in the digital twin took point of departure in the original design documentation of the observatory. This was adapted to represent the new state on base of the monitored data of the geometry of the timber elements. This data was as well queried to retain knowledge about details and inner composition of the laminated pieces of the original structure. The precise material setup of laminated beams was as such the base for the further design and structural simulation of the extension.

3.1.3 *Structure scale – adapting to availability*

In the third probe, ReShelter, we investigate how timber from a demolitioned structure with tectonic logics expressed in joints and cutouts in the reclaimed elements, can be adapted to an entirely new structural topology. ReShelter is a forest shelter made entirely from reclaimed timber of various sources. The main load bearing structure is a re-composition of reclaimed curved glue laminated beams produced for our 2021 RawLam project (Svilans et al. 2023), that from the limited stock of elements to form the central bifurcating arch of ReShelter from. An initial design of this load bearing arch uses key elements from this stock and provides an initial discretisation of the overall arch geometry. This rough fit is iteratively adapted by geometric operations, where individually elements, where systematically are moved and turned. When gaps and misalignments between adjacent elements remain after the first allocation run, the element with the largest differences is replaced in a second run with another from the remaining stock.

The shelter includes a timber platform made from new timber resource as well as rafters that interface with the arch. During the design, a structural assessment showed, that the load bearing capacity of the original RawLam joint system was not sufficient to take the higher loads, that arise from the exposed outdoor conditions. This required us to develop a new set of joints and to adapt the initial structural system further.

The adaptation of the RawLam structure to ReShelter capitalises on the malleability of the timber, where CNC milling allows to adapt the shape and cut interfaces. A problem with this approach is the unknown geometry of the reclaimed timber. Warp, local shrinkage and cracks over time results in deviations from the designed and fabricated geometries. Traditional wood

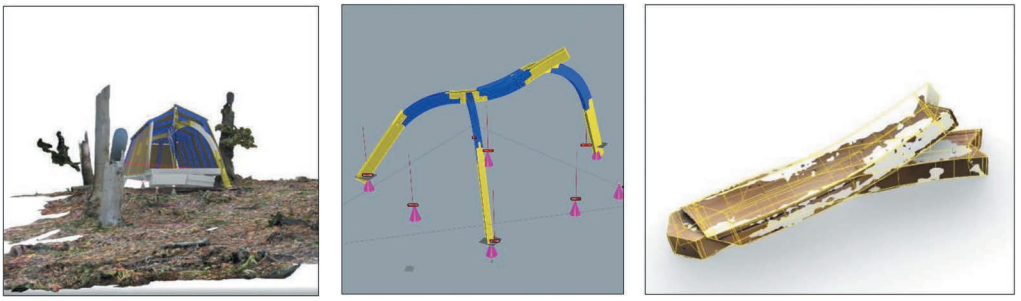


Figure 6. (from left) design of ReShelter positioned in a scan of the forest in hvalsø/denmark, The arch – the main structural system with new joint system, digital twin of a 3d scanned branching element used in reShelter.

craft uses preparatory cuts to conform the timber to e.g. rectangular geometries, that allows them to be further processes with precision with standard timber processes, such as saws and planers. This approach is waste, time and energy intensive and makes the reuse of timber in current practice often inefficient. In the case of the curved reclaimed timber in stock for ReShelter material, we found that such preparatory cuts with the 5-axis CNC machine would not take a lot of time but would as well reduce substantially the diameter and as such the load bearing capacity of the elements. In order to allow for a resource saving reclamation of the RawLam elements we develop a novel approach to the fabrication of timber elements, that takes point of departure in the monitoring of the found geometries and utilises the capacity of 5-axis CNC timber machinery.

In this process we 3D scan the reclaimed elements and use the monitored geometries to drive the design and fabrication instead of the original design specification and fabrication files. As in the Observatory probe, we find, that the monitored elements, deviate substantially due to fabrication tolerances, where the actual curvature of the beams does not match the specified ones, as well as warping during the first life span.

The design workflow in ReShelter includes finally feedback loops from structural analysis and fabrication level. These assure, that the overall adapted structure and joints have sufficient load bearing capacity, the machine at the fabricator can cut the elements and the interfaces with dependent building elements are working, while the overall design goals on e.g. functional level can be achieved. A computational workflow allows reuse timber elements directly without preparatory cuts and minimise in this way the cutting operations to the interfaces to other elements.

## 4 CONCLUSION: TOWARDS A BUILDING ECOLOGY

This paper has demonstrated that adaptation in circular timber construction requires distinct methodologies across material, element, and structure scales. By employing advanced monitoring and analysis techniques, reclaimed timber can be reintegrated effectively into sustainable design workflows. At the material scale, the *ReBeam* probe demonstrates how non-destructive monitoring techniques, such as CT scanning, enable detailed evaluation of reclaimed timber's heterogeneous properties, allowing for performance-driven assembly of glulam elements. At the element scale, the *Observatory* probe reveals how digital twin technologies and adaptive design strategies can accommodate structural changes and deformation in timber components, facilitating seamless extensions and reuse. At the structural scale, the *ReShelter* probe addresses the challenges of designing with constrained resource stocks, showcasing workflows that integrate reclaimed components into complex interdependent systems using iterative digital feedback loops. Across all three scales—material, element, and structure—the process of monitoring, analysis, and evaluation remains central but is tailored to each scale with different tools and methodologies.

The research presented in this paper expands the theoretical understanding of adaptability by reconceptualizing buildings as ecological systems embedded within and interacting across

networks of material cycles. This metabolistic perspective poses profound questions about architecture and its design and production. By highlighting the interconnectedness between the life spans of buildings and the material cycles flowing through them, it provides an architectural method for understanding how materials are integrated, reused, and repurposed within the building body. Rather than isolated structures, buildings are understood as vibrant ecosystems that respond to change while impacting and being shaped by the metabolic cycles of their materials.

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## AUTHOR CONTRIBUTIONS

The manuscript was written with the contribution of all authors. All authors have approved the final version of the manuscript. **Ramsgaard Thomsen, M.,** project conceptualization, methodology, design concept, writing – original draft, reviewing and editing, supervision, funding acquisition, **Dalager Nielsen, S.** design concept, writing – original draft, reviewing and editing, **Svilans, T.,** design concept, visualization, prototyping, fabrication, ideation - original draft, **Ee Pin Choo,** design concepts, visualization, prototyping, **Tamke, M.** project conceptualization, methodology, design concept, writing – original draft, reviewing and editing, supervision.

## REFERENCES

Harte, A.M. Mass timber – the emergence of a modern construction material. *Journal of Structural Integrity and Maintenance* 2(3):121–32.

Ormarsson, S., & Dahlblom, O. 2013. Finite element modelling of moisture related and visco-elastic deformations in inhomogeneous timber beams. *Engineering Structures* 49:182–9.

Ottenhaus, L.M., Yan, Z., Brandner, R., Leardini, P., Fink, G. & Jockwer, R. 2023. Design for adaptability, disassembly and reuse – A review of reversible timber connection systems. *Construction and Building Materials* 400:132823.

Hamida, M.B., Jylhä, T., Remøy, H. & Gruis, V. 2023. Circular building adaptability and its determinants – A literature review. *IJBPA* 41(6):47–69.

Giorgi, S., Lavagna, M. & Campioli, A. 2020. Circular Economy and Regeneration of Building Stock: Policy Improvements, Stakeholder Networking and Life Cycle Tools. In: Della Torre, S., Cattaneo, S., Lenzi, C., Zanelli, A. (eds.) *Regeneration of the Built Environment from a Circular Economy Perspective*.

McKinsey Sustainability. *Why the circular economy is all about retaining value* [Internet]. 2016. Available at: https://www.mckinsey.com/capabilities/sustainability/our-insights/why-the-circular-economy-is-all-about-retaining-value#/

Ghisellini, P., Cialani, C. & Ulgiati, S. 2016. A review on circular economy: the expected transition to a balanced interplay of environmental and economic systems. *Journal of Cleaner Production* 114:11–32.

Ramsgaard Thomsen, M., Miller, N. & Dalager Nielsen, S. *Building Ecology.* Ardeth. In print;15.

Cavalli, A., Cibecchini, D., TogniM. & , Sousa, H.S. 2016. A review on the mechanical properties of aged wood and salvaged timber. *Construction and Building Materials* 114:681–7.

Tamke, M., Svilans, T., Huber, J.A.J., Wuyts, W. & Thomsen, M.R. *Non-Destructive Assessment Of Reclaimed Timber Elements Using Ct Scanning – Methods And Computational Modelling Framework*.

Tamke, M., Gatz, S., Svilans, T. & Ramsgaard Thomsen, M. 2021. Tree-to-Product: Prototypical workflow connecting Data from tree with fabrication of engineered wood structure - RawLam. In: *Proceedings of WCTE202*: 2754–63.

Svilans, T., Choo, E.P., Ogoshi, M., Tamke, M. & Ramsgaard Thomsen, M. 2023. *Heterogenous Material Simulation In Architectural Timber Design*.